UNIVERSAL QUATIFIERS

Example #1

Let $P(x) : x+1 > x$ domain : all real numbers

What is truth value of $\forall_x P(x)$

Sol :

As $P(x)$ is true for all real numbers the quantification

$$\forall x \; P(x) \text{ is true}$$

The statement $\forall x \; P(x)$ is true if $P(x)$ is true for every x in D and false if $P(x)$ is false for at least one x in D

Example #2

Let $Q(x) : x < 2$ domain : all real numbers

Sol :

Here

$$\forall x \; Q(x) \text{ is false}$$

As at $x = 3$: $x < 2$

$3 < 2$ false

Here $Q(x)$ is false for $x = 3$ (Counter example)

An element for which $Q(x)$ is false is called Counter example of $\forall x Q(x)$

Example #3

Let $P(x) : x^2 > 0$ domain : all integers

Truth Value : ?

Sol :

Counter Example

$P(x)$ is false for $x = 0$, so

$$\forall x \; P(x) \text{ is false}$$

Example #4

Let $P(x): x^2 < 10$ domain: Positive integers not exceeding 4 $: \{1, 2, 3, 4\}$

Truth value = ?

Sol:

$\forall x\, P(x)$ is the same as

$P(1) \wedge P(2) \wedge P(3) \wedge P(4)$

$P(x)$ is false for $x = 4$ (Counter Example)

$P(4) : (4)^2 < 10$ false

So

$\forall x\, P(x)$ is false ($\forall x\, (x^2 < 10)$ is false) for domain $= \{1, 2, 3, 4\}$

When all elements in domain are listed, universal quantification $\forall x\, P(x)$ is the same as $P(x_1) \wedge P(x_2) \wedge \cdots \wedge P(x_n)$

Example #5

Let $P(x): x^2 \geq x$ where domain consists of all real numbers.

what is the truth value of

$\forall x\, (x^2 \geq x)$.

Sol:

Here inequality is false for all real numbers x with $0 < x < 1$.

For example:

$x = 1/2$

$P(1/2) : (1/2)^2 \geq 1/2$ false

So,

$\forall x\, (x^2 \geq x)$ is false

Example # 6

Let $P(x)$: if $(x > 1)$, then $(x+1 > 1)$ where domain = real numbers

What is the truth value of $\forall x\, P(x)$

Sol:

$P(x)$: if $(x > 1)$, then $x+1 > 1$

Here Hypothesis : $x > 1$

Conclusion : $x+1 > 1$

P	Q	$P \to Q$
T	T	T
T	F	F
F	T	T
F	T	T

Case 1:

if $x > 1$ is true (regardless of the specific value),

then

$x+1$ is greater than any value of x (and definitely $x+1 > 1$)

So, Conclusion is true for true hypothesis.

$\Rightarrow P(x)$ is true for all values of x

Case 2:

if $x > 1$ is true and we know that it could not be possible that $x+1 > 1$ to be false

if it is, then statement will be false.

Case 3 & 4: hypothesis = false Conclusion = T, F

if $x \leq 1$ i.e false hypothesis case

As x is real i.e $\{\cdots, -3, -\sqrt{2}, 0, 1, 4/5, 16, \cdots\}$,

then ~~P(x) is true~~ $x+1 > 1$ could be true ~~of~~ or false.

So

$P(x)$ is true regardless of whether the Conclusion is true or false.

So $\forall x\, (\text{if } (x > 1), \text{then } (x+1 > 1))$ is true.

Example # 7 (Restricted domain Case)

Express the statement using predicate and quantifier and then find its truth value.

$\forall$ quantification

Square is +ve

" For every real number x with $x < 0$, $x^2 > 0$ "

Sol:

$\underbrace{\text{For Every}}_{}$ $\underbrace{\text{real number } x}_{\text{domain}}$ with $\underset{\text{restriction}}{x<0}$, $\underbrace{\text{its square will be positive}}_{\text{Propositional Function}}$

$$\forall x < 0 \; (x^2 > 0)$$ where domain is all real number < 0.
−ve real numbers

As square of every −ve real number is positive, So

$$\forall x < 0 \; (x^2 > 0)$$

It Can also be written as:

$$\forall x \; (x < 0 \rightarrow x^2 > 0)$$

Example #8 (Restricted domain)

Express the following English statement into ~~the~~ logical expression:

" For every real number y with $y \neq 0$, we have $y^3 \neq 0$ "

or " The Cube of every nonzero real number is nonzero "

Sol:

$$\forall y \neq 0 \; (y^3 \neq 0) \quad \text{or} \quad \forall y \; (y \neq 0 \rightarrow y^3 \neq 0)$$

this is true as

the Cube of every nonzero real number is nonzero.

TRANSLATING FROM ENGLISH INTO LOGICAL EXPRESSIONS

Example #9

Express the following statement using Predicate and Quantifiers

"Every Student in the Class has studied Calculus"

Sol:

To identify the appropriate quantifiers, rewrite the statement as:

"$\underbrace{\text{For every}}_{\forall}$ student x $\underbrace{\text{in the class}}_{D}$, $\underbrace{x \text{ has studied Calculus}}_{C(x)}$".

Let $C(x)$: x has studied Calculus

domain : Students in the class

So, it can be expressed as :

$$\forall x \, C(x)$$

$\boxed{\text{Second Way:}}$ using wider domain

Rewrite the Statement as :

$\overbrace{\text{For every}}^{\forall}$ $\overbrace{\text{person } x}^{\text{Domain}}$, if $\overbrace{\text{person } x \text{ is a student in the class}}^{\text{hypothesis}}$, then $\underbrace{x \text{ has studied Calculus}}_{\text{(Conclusion)}}$.

$S(x)$: Person x is a Student in the class

$C(x)$: x has studied Calculus

domain : all people

it can be expressed as

$$\forall x \,(S(x) \rightarrow C(x))$$

Cannot be written as $\forall x\,(S(x) \wedge C(x))$ means all people are students in the class and has studied calculus

$\boxed{\text{Third way:}}$ 2-variable Quantifier

$Q(x, y)$: Student x has studied Subject y

domain : students in the Class

can be expression as

$$\forall x \, Q(x, \text{Calculus})$$

1st way:

$\forall x \, \underline{Q(x)}$ replace

$\forall x \, Q(x, \text{Calculus})$

2nd way:

$\forall x \, (S(x) \rightarrow \underline{Q(x)})$

$\forall x \, (S(x) \rightarrow Q(x, \text{Calculus}))$

Example # 10

Express the following english statement into Logical Expression.

"Every student in the class has visited Mexico or Canada"

Sol:-

To identify the appropriate quantifiers, rewrite the statement as:

"For every student x in the class, x has visited visited Mexico or x has visited Canada"

$C(x)$: x has visited Canada

$M(x)$: x has visited Mexico

domain : Student in the class

Can be expressed as

$$\forall x \, (C(x) \vee M(x))$$

$\boxed{\text{2nd Way :}}$ using wider domain

$S(x)$: x is student in the class

$C(x)$: x has visited Canada

$M(x)$: x has visited Mexico

domain : all people

$$\forall x \, (S(x) \rightarrow C(x) \vee M(x))$$

Can not be expressed as

$\forall x \, (S(x) \wedge (C(x) \vee M(x)))$

$\boxed{\text{Third way:}}$ 2-variable Quantifier

$V(x, y)$: $\overset{\text{student}}{x}$ has visited y Country

domain : Students in the Class

Can be expressed as :

$$\forall x \left(V(x, \text{Mexico}) \ \text{[illegible]} \ V(x, \text{Canada}) \right)$$

EXISTENTIAL QUANTIFIER

Example#1

Let $P(x): x > 3$ domain : all real numbers

what is the truth value of $\exists x\, P(x)$

Sol:

$P(x): x > 3$

The statement $\exists x\, P(x)$ is true if $P(x)$ is true for atleast one x in D and false if $P(x)$ is false for every value x in D

At least one x in domain is required to decide existential quantification.

At $x=4$ $P(4): 4 > 3$ true

Similarly $P(5): 5 > 3$ true

So,

$\exists x\,(x > 3)$ is true

Example #2

Let $Q(x): x = x+1$

domain : all real numbers

Truth value $\exists x\, Q(x) = ?$

Sol:

$Q(x): x = x+1$

$Q(x)$ is false for every real # x in D.

So, $\exists x\,(x+1)$ is false

Example #3

Let $P(x) : x^2 > 10$

domain: Positive integers not exceeding 4 $= \{1, 2, 3, 4\}$

Truth value $\exists x\, P(x) = ?$

Sol:

$P(x) : x^2 > 10$

$$\underset{F}{P(1)} \vee \underset{F}{P(2)} \vee \underset{F}{P(3)} \vee \underset{T}{P(4)} = T \qquad (P(4) : 16 > 10)$$

When all elements in domain are listed then $\exists x\, P(x)$ is the same as $P(x_1) \vee P(x_2) \vee \cdots \vee P(x_n)$

only element is sufficient to decide extentrial q.

So, $\exists x\, (x^2 > 10)$ is true

Example #4 (Restricted domain Case and Translation Case)

Express the following into logical expression:

$$\text{" }\underbrace{\text{There is a}}_{\exists}\ \underbrace{\text{real number } z \text{ with } z > 0}_{\text{Restricted Domain}}\text{, Such that } \underbrace{z^2 = 2}_{\text{Proposition}}\text{ "}$$

it can be expressed as:

$Q(z) : z^2 = 2$

domain: real numbers with $z > 0$

$$\exists z{>}0\, Q(z) \quad \text{or} \quad \exists z > 0\, (z^2 = 2) \quad \text{or} \quad \exists z\, (z > 0 \to z^2 = 2)$$

Truth value:

As domain is real with $z > 0$

$R = \{\cdots, -3, -\sqrt{2}, -\tfrac{1}{2}, 0, \tfrac{1}{2}, \sqrt{2}, \tfrac{4}{5}, \cdots\}$

for $x = \sqrt{2}$ $\quad Q(\sqrt{2}) : (\sqrt{2})^2 = 2$ true

So $\exists z > 0\, (z^2 = 2)$ is true

Translation From English into Logic Expression

Example # 5

Express the following statement into Logical Expression:

" Some Students in the Class has visited Mexico"

Sol:-

To identify the appropriate quantifiers, rewrite the above statement:

" There is a student x in this class having the property that x has visited Mexico."

it Can be expressed as

$M(x)$: x has visited Mexico

Domain : Students in the Class

So,

$$\exists x \, M(x)$$

2nd way: In a wider domain

"There is a person x having the property that x is a student in the Class and x has visited Mexico."

$S(x)$: x is a student in the Class

$M(x)$: x has visited Mexico

domain: all people

So, expression becomes:

$$\exists x \left(S(x) \wedge M(x) \right)$$

Canot be expressed as:

$$\exists x \left(S(x) \rightarrow M(x) \right)$$

because it is also true when

There is someone not in the Class.

F T | T

Trivial Case

NEGATING QUANTIFIED EXPRESSIONS

De Morgan's Laws for Quantifiers

$$\neg \forall x\, P(x) \equiv \exists x\, \neg P(x) \quad \text{—— (A)}$$

and

$$\neg \exists x\, P(x) \equiv \forall x\, \neg P(x) \quad \text{—— (B)}$$

Explanation

Consider the first equivalence regarding negated quantified expression

$$\neg \forall x\, P(x) \equiv \exists x\, \neg P(x)$$

Consider the following english statement :

"Every student in your class has taken a course in Calculus"

It Canbe expressed logically :

$\forall x\, P(x)$ where $P(x)$: x has taken a Course in Calculus

domain : Students in your Class

Negation of $\forall x\, P(x)$ Canbe expressed as :

$$\neg \forall x\, P(x)$$

i.e It is not the case that

every student in your Class has taken a Course in Calculus

The above negating statement canbe expressed using other quantification (Keeping negative sense) (Some, there is form : $\exists$)

"There is a student in your Class who has not taken a Course in Calculus"

As $P(x)$: x has taken a Course in Calculs

$\neg P(x)$: x has not taken a Course in Calculus

domain : Students in your class

Now the above english statement can be expressed as :

$$\exists x\, \neg P(x)$$

So, we Conclude that $\neg \forall x\, P(x) \equiv \exists x\, \neg P(x)$

Now Consider the 2nd Demorgan's Law

$$\neg \exists x \, P(x) \equiv \forall x \, \neg P(x)$$

Consider the following english statement

"There is a student in this class who has taken a Course in Calculus"

It Canbe expressed Logically:

$\exists x \, P(x)$ where $P(x)$: x has taken a Course in Calculus
domain: students in this Class

Negation of $\exists x \, P(x)$ Canbe expressed as:

$$\neg \exists x \, P(x)$$

i.e., "It is not the Case that
there is a Student in this class who has taken a course in Calculus"

The above negating Statement Canbe expressed using other quantification (for all, For every form) i.e. $\forall$
(Keeping negating sense)

"Every student in this class has not taken a Course in Calculus"

As $P(x)$: x has taken the Course in Calculus
$\neg P(x)$: x has not taken a Course in Calculus
domain: students in your Class

Now the above english statement Canbe expressed as:

$$\forall x \, \neg P(x)$$

So, we Conclude that

$$\neg \exists x \, P(x) \equiv \forall x \, \neg P(x)$$

Another Example

Example 2(a): Show that $\neg \exists x H(x) \equiv \forall x \neg H(x)$

"There is an honest politician"

$\exists x\, H(x)$ where $H(x)$: x is honest domain : all politician

$\neg \exists x\, H(x)$:

"It is not the Case
there is an honest politician"

Equivalent of $\neg \exists x\, H(x)$: (using other quantification keeping -ve sense)

Every politician is dishonest

or All politicians are not honest

$H(x)$: x is honest

$\neg H(x)$: x is not honest

$\forall x \neg H(x)$

So $\neg \exists x\, H(x) \equiv \forall x \neg H(x)$

Example 2(b): Show that $\neg \forall x\, C(x) \equiv \exists x \neg C(x)$

"All Americans eat Cheeseburgers"

$\forall x\, C(x)$ $C(x)$: x easts Cheeseburgers domain : all American

$\neg \forall x\, C(x)$:

"It is not the Case that
All Americans eat Cheeseburgers"

Equi "There is an American who does not eat cheeseburgers"

or "Some Americans does not eat cheeseburgers"

$\neg C(x)$: x does not eat Cheeseburger

$\exists x \neg C(x)$ So, $\neg \forall x\, C(x) \equiv \exists x \neg C(x)$

Examples From Mathematics:

Example :

Find the negation of the following:

(a) $\forall x\,(x^2 > x)$

(b) $\exists x\,(x^2 = 2)$

Sol:

(a) We have $\forall x\,(x^2 > x)$ and we have to find $\neg\forall x\,(x^2 > x)$

$$\neg\forall x\,(x^2 > x) \equiv \exists x\,\neg(x^2 > x) \quad \text{: By De Morgan's Law}$$

$$= \exists x\,(x^2 \leq x) \quad \text{Truth values depend on the domain}$$

(b) we have $\exists x\,(x^2 = 2)$ and we have to find $\neg\exists x\,(x^2 = 2)$

$$\neg\exists x\,(x^2 = 2) \equiv \forall x\,\neg(x^2 = 2) \quad \text{: By De Morgan's Law}$$

$$\equiv \forall x\,(x^2 \neq 2)$$

Example : Show that

$$\neg\forall x\,(P(x) \rightarrow Q(x)) \equiv \exists x\,(P(x) \wedge \neg Q(x))$$

Sol:

$$\neg\forall x\,(P(x) \rightarrow Q(x)) \equiv \exists x\,\neg(P(x) \rightarrow Q(x)) \quad \text{: Using DeMorgan's Law}$$

$$\equiv \exists x\,\neg(\neg P(x) \vee Q(x)) \quad : (\text{As } P \rightarrow Q = \neg P \vee Q)$$

$$\equiv \exists x\,\neg(\neg P(x)) \wedge \neg Q(x)) \quad : (\text{As } \neg(P \vee Q) = (\neg P \wedge \neg Q))$$

$$\equiv \exists x\,(P(x) \wedge \neg Q(x))$$

Remarks:

$\neg\forall x\,P(x)$ is the same as $\neg(P(x_1) \wedge P(x_2) \wedge, \dots, \wedge P(x_n))$ or $\neg P(x_1) \vee \neg P(x_2) \vee, \dots \vee \neg P(x_n)$

$\neg\exists x\,P(x)$ is the same as $\neg(P(x_1) \vee P(x_2) \vee, \dots, \vee P(x_n))$ or $\neg P(x_1) \wedge \neg P(x_2) \wedge, \dots \wedge \neg P(x_n)$

Show that $\neg \forall x\, P(x)$ and $\exists x\, \neg P(x)$ are logically equivalent no matter

- What the Propositional function is and
- What the domain is

$\neg \forall x\, P(x)$ is true iff $\forall x\, P(x)$ is false

$\forall x\, P(x)$ is false iff there is an element for which $P(x)$ is false

The statement

There is an element for which $P(x)$ is false holds iff there is an element for which $\neg P(x)$ is true.

There is an element for which $\neg P(x)$ is true iff $\exists x\, \neg P(x)$ is true.

putting things together:

$\neg \forall x\, P(x)$ is true iff $\exists x\, \neg P(x)$ is true

$\leftrightarrow$

it follows,

$\neg \forall x\, P(x)$ and $\exists x\, \neg P(x)$ are logically equivalent no matter what the propositional function is and what the domain is.

LOGICAL EQUIVALENCE INVOLVING QUATIFIERS

$$S \equiv T$$

Here S and T are statements involving predicates and quantifiers. S and T are logically equivalent iff they have same truth table, no matter

- which predicates are substituted into these statements

and

- which domain is used for the variables in these propositional functions

1. $\forall x \left(P(x) \land Q(x)\right)$ and $\forall x\, P(x) \land \forall x\, Q(x)$ are logical equivalent

2. $\exists x \left(P(x) \lor Q(x)\right)$ and $\exists x\, P(x) \lor \exists x\, Q(x)$ are logical equivalent.

1. implies
We can distribute a Universal quantifier over a Conjunction
(we cannot distribute a Universal quantifiers over a disjunction)

2. implies
We can distribute an existential quantifier over a disjunction.
(we cannot distribute an existential quantifier over a conjunction)

For Proof : Visit Book Page 45

PRECEDENCE OF QUATIFICATION

$\forall$ and $\exists$ have higher precedence than all logical operators from Propositional Calculus.

For example,

$\forall x\, P(x) \lor Q(x)$ is the same as $\left(\forall x\, P(x)\right) \lor Q(x)$

BINDING AND FREE VARIABLES

All variables that occur in a propositional function must be bound or set equal to a particular value to turn into a proposition.

Consider the Statement

$$\exists x\,(x+y=1)$$

Here variable x is bound by existential quantification $\exists x$ and y is free because it is not bound by a quantifier

SCOPE:

The part of a logical expression to wich a quantifier is appleed is called the scop of this quantifier.

Free Variable :

A variable is free if it is outside the scope of all ~~quantification~~ quantifiers in the formula that specify this variable.

Bound Variable :

When a quantifier is used on the variable x, we say that this occurrence of the variable is bound.

EXAMPLE :

$$\exists x\,(P(x) \wedge Q(x)) \vee \forall x\, R(x)$$

All variables are bound.

Existential quantifier $\exists x$ binds the variable x in $P(x) \wedge Q(x)$

Universal quantifier $\forall x$ bids the variable x in $R(x)$

Scope of $\exists x$ is in the expression $(P(x) \wedge Q(x))$

Scope of $\forall x$ is in the expression $R(x)$

Consider another statement $\exists x\,(P(x) \wedge Q(x)) \vee \forall y\, R(y)$

In this statement, scopes of two quantifiers donot overlap.